

# **Chapter 3**

## **System Analysis 1**

### **Fact Gathering**

# System Development Life Cycle (SDLC)

Phase 1

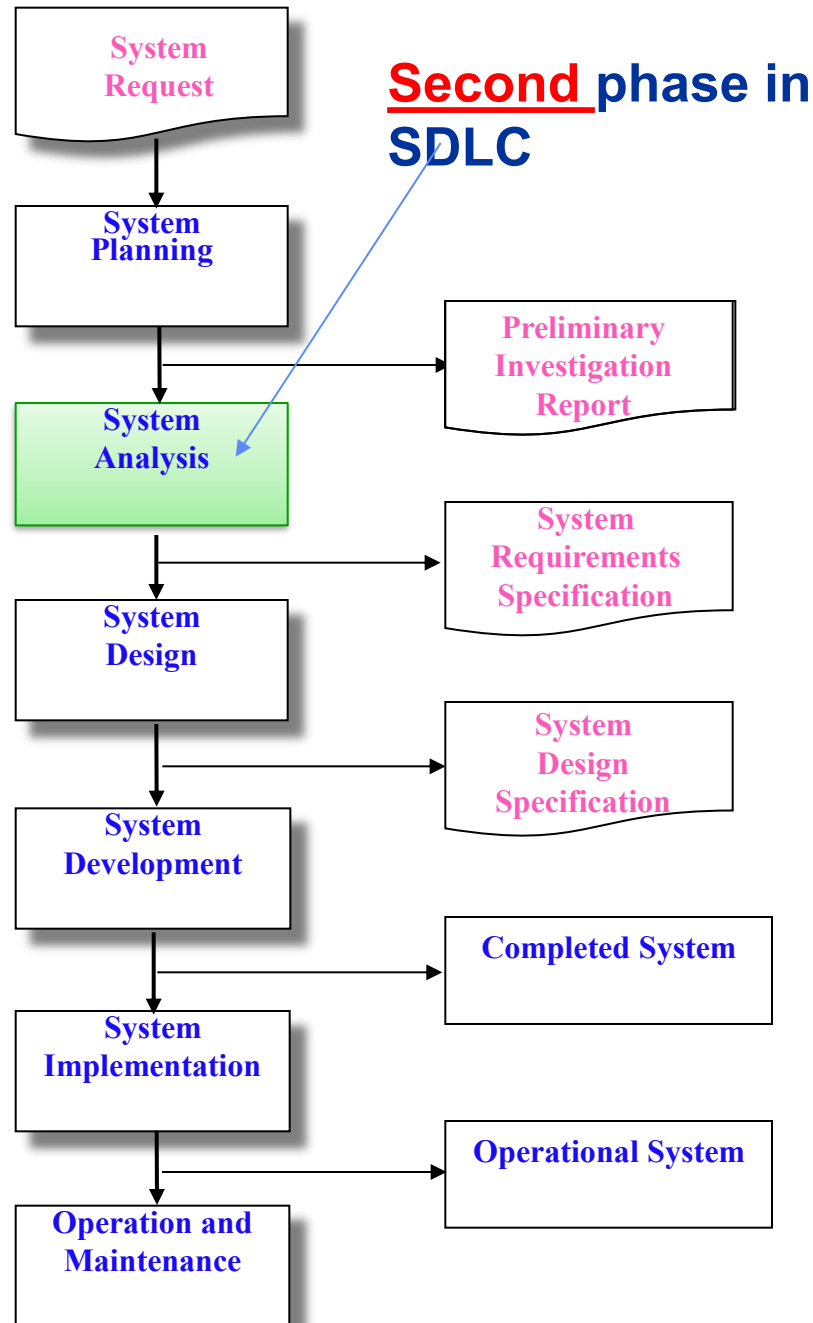
Phase 2

Phase 3

Phase 4

Phase 5

Phase 6



# Definition of System Analysis (Requirements Analysis)

- **Another name.** Sometimes called 'Requirements Analysis'.
- **Objectives.** Defining the required functionalities of the new system.
- **Document.** These requirements are usually collated in a list (**System Requirements Specification**).
- **Fact-Gathering Techniques.** The analyst will use fact-finding techniques such as interviewing, document analysis and questionnaires.
- **Fact-Recording Techniques.** The documented facts will be written and/or modelled using such techniques as narrative description, **DFD** and **ERD**.

# Fact Gathering in System Development

- **From whom.** From people who will be using the system ie. mainly from **end users**.
- **Quality of System Analyst (IT staff).** The system analyst must have *knowledge, skills* and be *experienced* in the business area eg. business rules and processes involved, and interviewing skills
- **What to gather.** Business *rules, processes, environment* of work, *activities* of users, and system *interfaces* with other systems, *inputs, outputs* and *data stores* (files).

# Techniques of Fact Gathering

# Fact Gathering

- **Methods** - many methods to gather facts (users' requirements) eg. interviewing
- **Department functions** – A company has many departments and each has its **department functions**.
- **System functions** – each system has its **system functions** used to support its business processes
- **Process** – a process consists of **activities** or tasks, carried out by system and/or people.

# 1. Interviews



- **Meaning**

- **Widely used** technique and the most productive.
- Ask the user **face-to-face** on how the current system works and about requirements.
- **Formal** meetings (with agenda, objectives).
- Documented **feedback** to the user for confirmation

# Advantages of Interviews

- **Overcome Resistance.** Provides an opportunity to meet and overcome user resistance. It gets cooperation of people involved.
- **Body language.** Interviews allow the system analyst (IT staff) to observe the tone of the responses and body language which convey great deal of information.
- **Clarify Facts.** Face-to-face interview allows the interviewer to verify and clarify the confusing facts.
- **Build Rapport.** Opportunity to develop rapport with users and to seek cooperation.
- **Intimate and Frankness (honesty) .** People who would be unwilling to put critical or controversial comments in writing , talk more freely in person.

# Disadvantages of Interviews

- **Costly and Time-consuming.** An interview requires the dedicated time of both the interviewer and the interviewee for the duration of the interview.
- **Require Skills.** Certain amount of skill is required on the part of the interviewer and interviewing is an art not readily acquired. (communication skill, observation skill, patience )
- **Require Personal Contacts.** Personal contacts are important in getting the cooperation of the people involved.

## 2. Questionnaires

- **Meaning**

- Users are **sent** some questionnaires about the current system for them to answer at their leisure time.
- A **mix** of closed questions (to ascertain facts) and open questions (to describe feelings and opinions).
- Can be checked and **tested** before sending them to the users to respond.
- May represent a **first stage** of data collection to be supported by later follow-on interviews.

# Questionnaires

- **Circumstances Suitable**

- **For large number of users (respondents).** There may be insufficient time to interview all the possible users of the system and therefore a sample is taken.
- **For geographically distributed users.** It is time-consuming and expensive to interview users who are widely geographically distributed.
- **For sensitive issues.** Useful in circumstances where the provider of the information wants to remain anonymous for sensitive issues.
- **For facts gathering which takes time.** Factual information requires the users to investigate rather than respond with instant answers as in interviews.

# Example of Questionnaire

## PURCHASE REQUISITION QUESTIONNAIRE

Pat Kline, Vice President, Finance, has asked us to investigate the purchase requisition process to see if it can be improved. Your input concerning this requisition process will be very valuable. We would greatly appreciate it if you could complete the following questionnaire and return it by March 10 to Dana Juarez in information technology. If you have any questions, please call Dana at x2561.

### A. YOUR OBSERVATIONS

Please answer each question by checking one box.

1. How many purchase requisitions did you process in the past five working days? \_\_\_\_\_
2. What percentage of your time is spent processing requisitions?  
☐ under 20%                      ☐ 60–79%  
☐ 21–39%                        ☐ 80% or more  
☐ 40–59%
3. Do you believe too many errors exist on requisitions?  
☐ yes  
☐ no
4. Out of every 100 requisitions you process, how many contain errors?  
☐ fewer than 5                      ☐ 20 to 29  
☐ 5 to 9                                ☐ 30 to 39  
☐ 10 to 14                            ☐ 40 to 49  
☐ 15 to 19                            ☐ 50 or more
5. What errors do you see most often on requisitions? (Place a 1 next to the most common error, place a 2 next to the second, etc.)  
☐ incorrect charge number                      ☐ missing authorization  
☐ missing charge information                      ☐ other (please explain) \_\_\_\_\_  
☐ arithmetic errors  
☐ incorrect discount percent used

### B. YOUR SUGGESTIONS

Please be specific, and give examples if possible.

1. If the currently used purchase requisition form were to be redesigned, what changes to the form would you recommend?  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

(Please attach another sheet if necessary)

2. Would you be interested in meeting with an information technology representative to discuss your ideas further? If so, please complete the following information:

Name \_\_\_\_\_ Department \_\_\_\_\_

Telephone \_\_\_\_\_ E-mail address \_\_\_\_\_

# Example of Questionnaires

- Focus on a particular **issue or problem or objective**
- Inventory System
  - **Problem/Issue** : Lack of useful reports in the inventory system.
  - **Questions** :
    - What are the reports which are **produced** by the present system ?
    - What do you think are some of the reports which are **missing** but could be useful in your decision making ?
    - Describe each of the **reports** you would like to have: title, content, who use, purpose.
    - How would each of these reports, which you have suggested, **help** you in your decision making ?
    - Do functions and **features** in the system available allow you to create your own reports ?
    - Have you been **trained** to use the system, particularly on the reporting functions and features.
    - What do you think are the **reasons** why the system cannot produce the reports that you want ?

**NOTE** : Do not simply copy this example in assignment (plagiarism)

# Advantages of Questionnaires

- **Recipient Answer at Convenient Time.** The recipients can answer questionnaires at their convenience; it isn't necessary to set aside a significant block of time as one must do for an interview.
- **Respond Anonymously (namelessly) .** The questionnaire can be used to provide anonymous responses. There is evidence to suggest that respondents are more likely to give more accurate responses this way.
- **Respondent Given Time.** The respondent is given time to assemble/gather/collect the required information.
- **Gather Information from Large Group.** Questionnaires may be used where the respondents are relatively many and geographically dispersed.

# Disadvantages of Questionnaires

- **Difficult to Design.** Difficult to formulate or design the questions so that no misinterpretation is possible and no bias is possible in the replies.
- **Cannot Clarify Questions.** One cannot clarify the questions as you can in an interview.
- **Not all the Forms Will be Returned.** Many people object to filling in forms while others delay completing them until they are eventually forgotten.

# 3. Observation - Meaning



- **Objectives.** An analyst would **watch** who interrelates with whom in a depth and the procedures involved
- **Arrangement.** Make arrangements and make everyone aware of the purpose of your visit.
- **Items observed.** Observe how **data flows**, from desk to desk or computer to computer. Note how data comes into and leaves the organisation.
- **Method.** Participatory observation; in this method the system analyst temporarily joins the activities of the group.

# Advantages of Observation

- **Additional Perspective.** Seeing the system in action gives you an additional perspective to supplement what you have heard (interview) and read (inspection).
- **Environmental Conditions.** Allows the analyst to observe environmental conditions (such as dirt, space and noise) which may affect implementation.
- **Cross-check and Verify.** Provides the opportunity for the analyst to cross-check and verify information given in an interview.
- **Acquire New Know-how.** Personal observation helps you acquire the know-how you will need for implementation.
- **Better Acceptance.** You also become better acquainted with the operating personnel who will be implementing the new or changed system.

# Disadvantages of Observation

- **Time Consuming.** Due to time factor, the analyst seldom carry out thorough investigation.
- **Need Prior Understanding of Procedure.** It needs the observer to have an understanding of the procedures involved for effective observation.
- **Need Power of Concentration.** Good observation is very difficult to develop. Much depends on power of concentration. Easily distracted.
- **Subconscious Observation.** An analyst will tend to make subconscious observations of his local environment and based on past experiences.
- **Hawthorne Effect.** Workers might be nervous under observation. Therefore may not accurately represent the real situation.

# 4. Analysis of Existing Documents

- **Collect.** The analyst will collect a number of relevant documents used in the current system.
- **Copy.** Make copies or samples of relevant documents. Examples : reports, organisation charts, procedures, manuals etc.
- **Analyse.** Make analysis/study of the documents to understand the problems and processes.



# 5. Joint Application Development (JAD)

- **Forum.** JAD is a *workshop* for users in exploring requirements, issues and problems.
- **Purpose.** Allows IT personnel to work with users as a group to learn about their requirements through *discussion*.
- **Participants.** Headed by a project leader who has various skills eg. organisational, interpersonal, technical (eg CASE tools)
- **Documentation.** Proceedings of the workshop are recorded.

# 6. Background Research

- **When to Use :**

- There are **no expertise** available for interview or questionnaires
- No **existing** system and documents available for observation or inspection.

- **Methods :**

- Reviewing **journals**, periodicals, and books that contain information relevant to the task at hand.
- Attending professional **meetings** and seminars.
- **Formal** and informal discussions with other professional in related areas.
- **Visits** to similar sites.

# 7. Prototyping Approach

- **Model.** A model containing key features eg data entry, or report program, is created by using a 4GL or CASE tools.
- **Purpose.** For users to test and provide feedback as to whether it meets their requirements.
- **Speed.** The prototypes can be modified quickly eg using 4GL or CASE tools.
- **Productivity.** The models or prototypes can be developed into a working and complete application programs in shorter time period.

# Prototyping Process

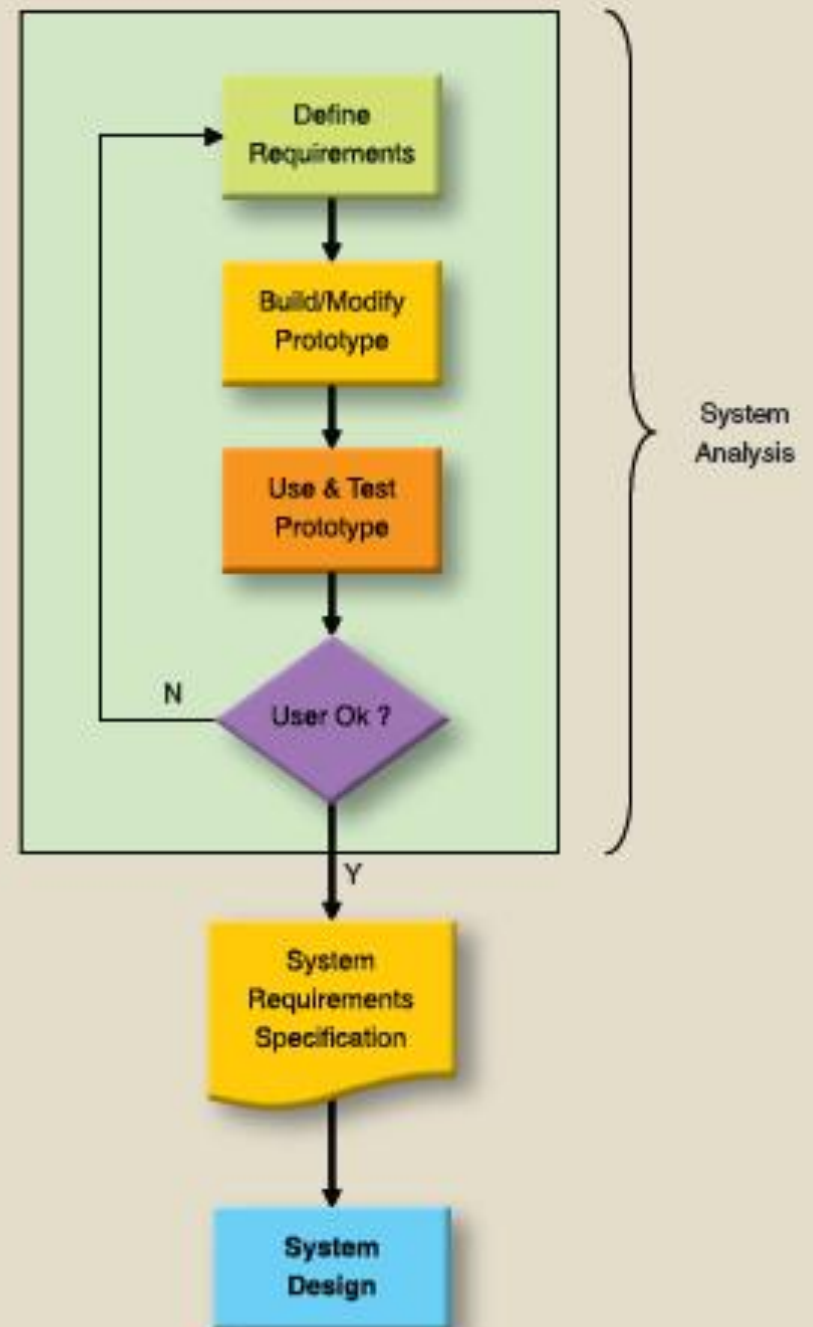
**Users and IS  
Personnel**

**IS  
Personnel**



**Users**

# Steps of Prototyping Process



# Exercise for Students

- **KL Capital Management helps to manage RM5 billion ( RM5,000,000,000) in assets for 200 clients. Currently KL Capital employs 20 consultants without the use of systems. The company is planning to implement expert system to assist the consultants.**
- **Required**
  - **Identify and briefly describe 2 fact-gathering methods to be used.**
  - **Justify your selection of the fact-gathering methods.**

# Solutions

- **Fact-gathering Methods : (any 2)**
  - Interviewing
  - Questionnaires
  - Research
  - Prototyping
- **Justification**
  - Analyst and users are new to the system
  - Users need to see prototypes before confirming
  - Analysts need to learn new knowledge

# Functional Requirements

# Functional Requirements

- Refers to the **functionalities** that the system is expected to provide eg create Purchase Order (for Purchasing System).
- Described in detail **how** the functionalities are used to support a business process, activities or tasks being carried out including inputs, process, outputs and stored data.

# Example of Functional Requirements

## – Sales System

- **Enquiries**
  - Receive enquiries (UK) /inquiries (US)
  - Answer enquiries
- **Quotations**
  - Receive RFQ (Request for Quotation/Price)
  - Create and send Quotations

**Note :** the above are just very brief descriptions (key words) of the functionalities.

# Non-Functional Requirements

Do not directly related to specific **functions** of the system.

Describe other features, constraints and characteristics of the system.

# **Example 1 : Interfaces with other systems**

- **With other systems** - there is an increasing need for applications to receive data from one system and to supply it to another.
  - **Example** - Payroll system with Accounts system for cash flow forecasting.
- **With generalized software** - such as spreadsheets or specialised tools for data mining.

## Example 2 : Audit requirements

- **Meaning** - most software solutions require some sort of **audit trail**. This trail records significant information about particular *transactions*.
- **Example** - payroll system : the change of salary scale for a particular employee might be recorded on an audit trail eg. date, time of change, the previous salary scale etc.
- **Agreement** - the need for, and content of, an audit trail must be agreed with auditors and defined in the non-functional requirements.

# Example 3 : Legal issues

- **Importance** - the software may have to support certain legal requirements. These may be defined by industry-wide legislation, such as the Data Protection Act (in UK).
- **Purpose** - legislation may place certain responsibilities on the **user**; the software should support the user in fulfilling those responsibilities.
- **Example** - the legislation may require fields recording who provided that information and when that information was last updated etc

# Example 4 : Archiving, backup and recovery

- **Backup and Recovery Routines**

- **Definition** - to make a copy in anticipation of future failure or corruption.
- **Purpose** - to ensure that the most recent usable copy of the data can be recovered and restored in the event of loss or corruption.
- **Method** – manually or automated using software (part of OS function)
- **Intervals for Back-up** - depending on (1) nature of data and (2) nature of organization

- **Archiving Facilities**

- **Definition** - the process of moving (by copying) data from primary storage, such as a hard disk, to tape or other portable media for long-term storage.
- **Purpose** - provides a legally acceptable business history, while freeing up hard disk space, speed up searching function.
- **Length of Retention** depends on : (1) Legal obligations and (2) Other business needs.

# Example 5 : Usability

- **Usability** – ease with which end users can use a particular system to perform a particular business function.
- **Methods** used to measure :
  - **Learnability** – How long it takes a user to learn to use a particular function of the system ?
  - **Memorability** – When users return to the design system after a period of not using it, how easily can they remember how to use ?
  - **Satisfaction** – How pleasant is it for the users to use the design system ?
  - **Errors** – How many errors do users make, how severe are these errors, and how easily can they recover from the errors?
  - **Efficiency** – Once users have learned the design system, how quickly can they use the system to perform tasks ?

# Example 6 : Reliability

- Measures

- System's ability to perform and continue to perform its functions without **failure**.
- Systems which are not reliable will **break down** frequently.

- Consequences

- System out of service brings hardship to the users and others eg. customers.

- Reasons for not being reliable

- Insufficient testing
- Poor design

- Important in some industries

- Life-support systems
- Attack and defend systems

# Exercise for students

- **Inventory System :**
  - Describe 2 **functional** requirements
  - Describe 2 **non-functional** requirements

# Solutions

- Describe 2 functional requirements
  - Recording goods received from suppliers
  - Deliver goods to customers.
  - Transfer goods between warehouses.
- Describe 2 non-functional requirements
  - User-friendly features eg. graphical user interface (GUI), use of RFID
  - Provide audit trails for the auditors
  - Increase the reliability of the system so that it would not fail frequently

Food point of sale program/ food order program processes and records the customer food order. The system use graphical user interface (GUI)

REG # 1 RON W. -- Plexis Point Of Sale

**SALE TOTAL**  
**20.58**

CUST OPS - F8

**TAKE OUT**

F-11 F-12 F2 - ITEMS

| DESCRIPTION           | QTY | TOTAL | T |
|-----------------------|-----|-------|---|
| CHICKEN CLUB SANDWICH | 1   | 5.99  | T |
| PIZZA SLICE           | 1   | 1.99  | T |
| TURKEY BURGER         | 1   | 4.99  | T |
| TURKEY CLUB SANDWICH  | 1   | 5.99  | T |
| MUSTARD               | 1   | 0.25  |   |
| MAYO                  | 1   | 0.00  |   |

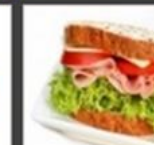
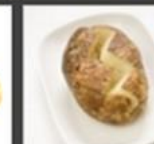
GFT HLD D-IN

DISC CPN TCKTS

TIPS  DWR

MAIN MENU

LUNCH MENU NO PICTURES SAMPLE DINNER MENU BREAKFAST MENU DRINKS MUSTARD KETCHUP MAYO PICKLES ETC

|                                                                                     |                                                                                     |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|    |    |    |    |    |    |
|    |    |    |    |    |    |
|   |   |   |   |   |   |
|  |  |  |  |  |  |
| TAXABLE ITEM                                                                        | NON TAXABLE ITEM                                                                    |  |  |  |  |

F1 - PAD

F3 - CASH

F4 - CREDIT

F5 - PAY

F6 - CASE/PK

F7 ACTIONS

F10 OPTIONS

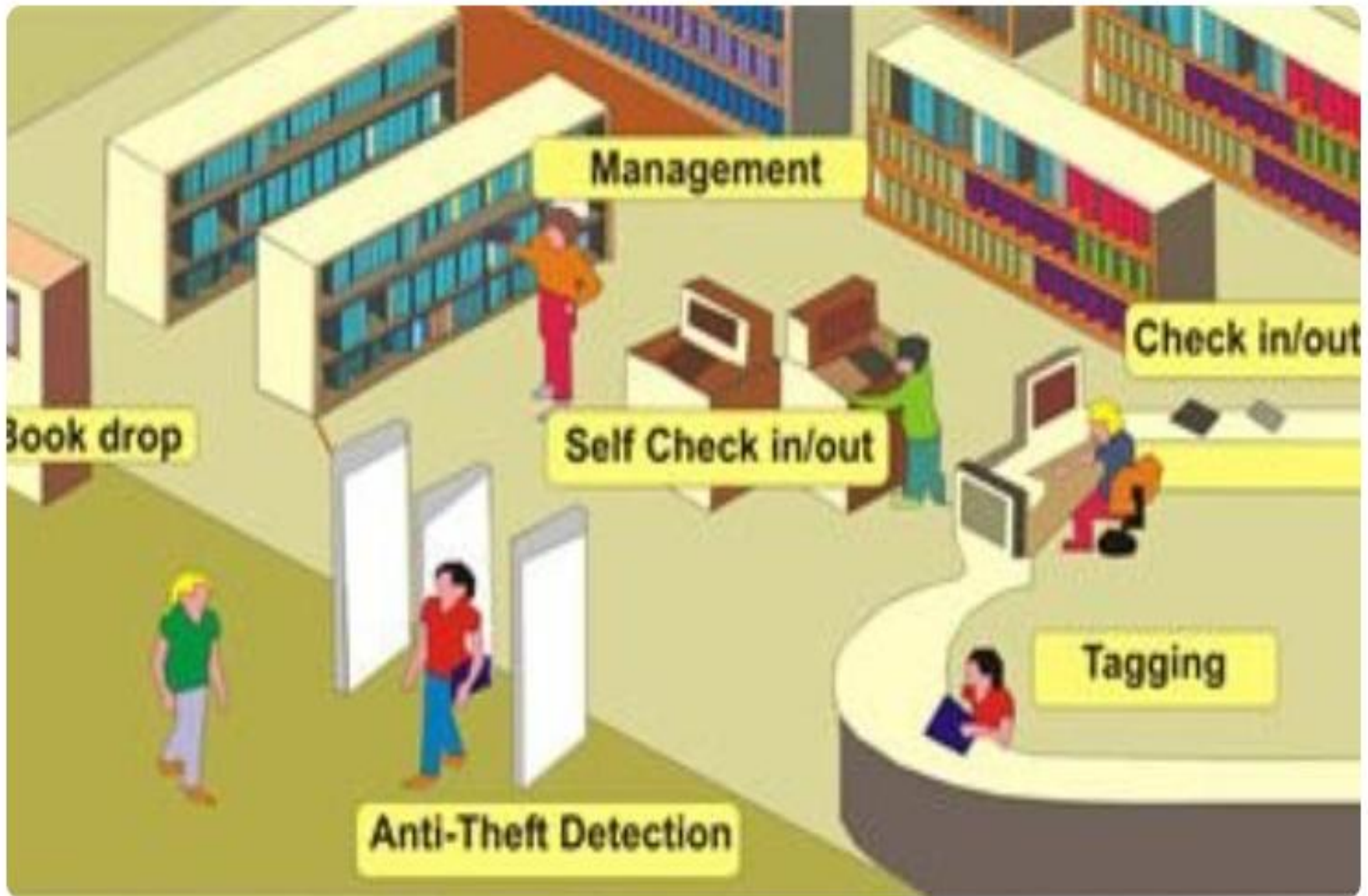
NON-TAX DISCOUNT TTL TAX SUBTOTAL

MARINA ROCKSIDE CAFE

5:10:42 PM

- <https://www.linkedin.com/pulse/integration-rfid-library-management-software-r-sunil-kumar>

## use of RFID



# use of RFID

